UIC Department of Chemical Engineering
UNIVERSITY OF ILLINOIS AT CHICAGO
COLLEGE OF ENGINEERING

ASSIGNMENT #3

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

Your name: Thomas Mazur
Due Date: Wednesday February 19 by 5:00pm in my office

Number of points..............................100
Your score................................~~73~~ 71

Note: All problems are worth 100 points. You final score will be the average value

Assumptions for system operation:

Heat transfer coefficient obtained from CSTR Aspen simulation
Density of mixture in CSTR remains constant
Density of cooling water remains constant in cooling jacket
Initial volume of mixture in CSTR is half of the total volume of the CSTR
Constant process flow
Well mixed reactor
Initial concentrations of reactants are the same values from HW 1
Initial Cooling water inlet temperature is 298.15 K

Lots of

white-space.

There is a lot of work missing that could have fit here.

Listed set points?

−9

2) Process Gain of feedback controller:

$$K_{P,FB}\ [=]\ \frac{\Delta°\mathrm{C}\ (Reactor\ Temp)}{\%\Delta(CO)}$$

ΔK	349-341	341-355	355-349
$\%\Delta(CO)$	-10	20	-10
$K_{P,FB}$	-0.8	0.7	-0.6

units

−2

3) Process Gain of cascade controller:

$$K_{P1,CS}\ [=]\ \frac{\Delta°\mathrm{C}\ (Cooling\ Jacket\ Temp)}{\%\Delta(CO)}$$

ΔK(CJ)	321.5-311.5	311.5-339	339-321.5
$\%\Delta(CO)$	-10	20	-10
$K_{P1,CS}$	-1.0	-0.375	-1.75

$$K_{P2,CS}\ [=]\ \frac{\Delta°\mathrm{C}\ (Reactor\ Temp)}{\Delta°\mathrm{C}(Cascade\ set\ point\ temp)}$$

ΔK(RT)	349-356.5	356.5-346	346-349
$\%\Delta(Set\ Point)$	-10	20	-10
$K_{P1,CS}$	0.75	0.525	0.3

Final values?

Eqn. labels.

−4

4) Process Gain of feed forward controller:

$$K_{P,FF} \,[=]\, \frac{\Delta°C\ (Reactor\ Temp)}{\Delta°C(Cooling\ water\ inlet\ temp)}$$

$\Delta K\ (RT)$	349-352	352-347	347-349
$\Delta K(CW)$ K	298.15-308.15	308.15-288.15	288.15-298.15
$K_{P,FF}$	0.3	0.25	0.2

Steady State Code:

```
!Variabe Declarations and Data
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
 Parameter Rg=8.3145 As Real  !Universal Gas Constant, J/(mol K)

 Global AAFeed, EtohFeed, WFeed, QF, Ua, Qcool,DF, MG, MMW, Fws As Real
 AAFeed = 66.7148 !mol/sec
 EtohFeed = 64.5584     !mol/sec
 WFeed = 3.50421  !mol/sec
 QF = 0.007570824 !m3/sec
 Ua = 21586.3     !J/K sec
 Qcool = 431726   !J/mol
 DF = 7.26        !PSI
 Fws = .006 !m3/sec

 MG = 64.172      !Gal to m3 conversion
 MMW = 18.02      ! mol weight of H2O


 Global kf0, kr0, Ef0, Er0 As Real
 kf0 = 1.9E+5     !m3/mol sec
 kr0 = 5E+4       !m3/mol sec
 Ef0 = 59500      !J/mol
 Er0 = 59500      !J/mol

 Global V, Tr, rho, Cp, DHr, Tfeed, Tw, Cpw, Vc, rhoW, rhoWM As Real
 V = 10.0         !m3
 Tr = 343.15      !K
 rho = 17000      !mol/m3
 Cp = 100.0315    !J/mol K
 DHr= -29115      !J/mol
 Tfeed = 298.15   !K
 Tw = 298.15      !K
 Cpw=  75.4       !J/mol K
 rhoW = 55388.9   !mol/m3
 Vc = 0.1*5       !kg/m3
rhoWM = 997      !kg/m3
```

```
@Initial Conditions
 U(1)=16902.02     !mol/m3
 U(2)=0.0
 U(3)=0.0
 U(4)=54845.05     !mol/m3
 U(5)= 323.15      !K
 U(6)= 298.15      !cooling water temp

@Model Equations
 Dim CAA, CEtoh, CEAC, CW, Temp, Tc As Real

 CAA = U(1)
 CEtoh = U(2)
 CEAC = U(3)
 CW = U(4)
 Temp = U(5)
 Tc = U(6)



 Dim kfor, krev, R1, Fcw As Real

 Fcw = 36*sqrt(DP)*MG*rhoW*(1/MMW)*(1000/60)
 kfor= kf0*exp(-Ef0/(Rg*Temp))
 krev= kr0*exp(-Er0/(Rg*Temp))
 R1 = kfor*U(1)*U(2) - krev*U(3)*U(4)


 !If(iRange==1) Fcw = 36*sqrt(DP)*MG*rhoW*(1/MMW)*(1000/60)
 !If(iRange==2) Fcw = 60.4*sqrt(DP)*MG*rhoW*(1/MMW)*(1000/60)
 !If(iRange==3) Fcw = 21.7*sqrt(DP)*MG*rhoW*(1/MMW)*(1000/60)
 !If(iRange==4) Fcw = 36*sqrt(DP)*MG*rhoW*(1/MMW)*(1000/60)
```

```
F(1)=AAFeed - QF*CAA  -R1*V
F(2)=EtohFeed - QF*CEtoh  -R1*V
F(3)= -QF*CEAC + R1*V
F(4)= WFeed - QF*CW + R1*V
F(5)=QF*rho*Cp*(Tfeed - Temp) - Ua*(Temp-Tc) - DHr*R1*V
F(6)=Fws*rhoW*Cpw*(Tw - Tc) + Ua*(Temp-Tc)

@Coefficient Matrix
E(1)=V
E(2)=V
E(3)=V
E(4)=V
E(5)=rho*Cp*V
E(6)=rhoW*Cpw*Vc


@Solver Options
Neq=6
Npts=80
Tend=8000
Tstop=50;100;150
Param(28)=1
VariableNames=Time(Sec);CAA;CEtoh;CEAC;CW;Temp(K);Tc(K)
```

Dynamic Feedback Code:

```
!Variabe Declarations and Data
!*******************************
 Parameter Rg=8.3145 As Real  !Universal Gas Constant, J/(mol K)

 Global AAFeed, EtohFeed, WFeed, QF, Ua, Qcool,DP, MG, MMW, Fws As Real
 AAFeed = 66.7148 !mol/sec
 EtohFeed = 64.5584     !mol/sec
 WFeed = 3.50421  !mol/sec
 QF = 0.007570824 !m3/sec
 Ua = 21586.3     !J/K sec
 Qcool = 431726   !J/mol
 DP = 7.26        !PSI
 Fws = .006       !m3/sec

 MG = 64.172      !Gal to m3 conversion
 MMW = 18.02      ! mol weight of H2O


 Global kf0, kr0, Ef0, Er0 As Real
 kf0 = 1.9E+5     !m3/mol sec
 kr0 = 5E+4       !m3/mol sec
 Ef0 = 59500      !J/mol
 Er0 = 59500      !J/mol

 Global V, Tr, rho, Cp, DHr, Tfeed, Tw, Cpw, Vc, rhoW, rhoWM As Real
 V = 10.0         !m3
 Tr = 343.15      !K
 rho = 17000      !mol/m3
 Cp = 100.0315    !J/mol K
 DHr= -29115      !J/mol
 Tfeed = 298.15   !K
 Tw = 298.15      !K
 Cpw=  75.4       !J/mol K
 rhoW = 55388.9   !mol/m3
 Vc = 0.1*5       !kg/m3
 rhoWM = 997      !kg/m3
```

```
@Initial Conditions
 U(1)=3.16305E+03
 U(2)=2.87822E+03
 U(3)=5.64904E+03
 U(4)=6.11190E+03  !mol/m3
 U(5)=3.49034E+02  !K
 U(6)=3.21699E+02  !cooling water temp


@Model Equations
 Dim CAA, CEtoh, CEAC, CW, Temp, Tc As Real

 CAA = U(1)
 CEtoh = U(2)
 CEAC = U(3)
 CW = U(4)
 Temp = U(5)
 Tc = U(6)


 Dim kfor, krev, R1 As Real

 !Fcw = 36*sqrt(DP)*MG*rhoW*(1/MMW)*(1000/60)
 kfor= kf0*exp(-Ef0/(Rg*Temp))
 krev= kr0*exp(-Er0/(Rg*Temp))
 R1 = kfor*U(1)*U(2) - krev*U(3)*U(4)


 If(iRange==1) Fws = .006
 If(iRange==2) Fws = .020
 If(iRange==3) Fws = .002
 If(iRange==4) Fws = .006
```

```
F(1)=AAFeed - QF*CAA  -R1*V
F(2)=EtohFeed - QF*CEtoh  -R1*V
F(3)= -QF*CEAC + R1*V
F(4)= WFeed - QF*CW + R1*V
F(5)=QF*rho*Cp*(Tfeed - Temp) - Ua*(Temp-Tc) - DHr*R1*V
F(6)=Fws*rhoW*Cpw*(Tw - Tc) + Ua*(Temp-Tc)

@Coefficient Matrix
E(1)=V
E(2)=V
E(3)=V
E(4)=V
E(5)=rho*Cp*V
E(6)=rhoW*Cpw*Vc


@Solver Options
Neq=6
Npts=80
Tend=8000
Tstop=1000;2000;3000;4000
Param(28)=1
VariableNames=Time(Sec);CAA;CEtoh;CEAC;CW;Temp(K);Tc(K)
```

Cascade Controller Code:

```
!Variabe Declarations and Data
!*******************************
 Parameter Rg=8.3145 As Real  !Universal Gas Constant, J/(mol K)

 Global AAFeed, EtohFeed, WFeed, QF, Ua, Qcool,DP, MG, MMW, Fws As Real
 AAFeed = 66.7148 !mol/sec
 EtohFeed = 64.5584     !mol/sec
 WFeed = 3.50421  !mol/sec
 QF = 0.007570824 !m3/sec
 Ua = 21586.3     !J/K sec
 Qcool = 431726   !J/mol
 DP = 7.26        !PSI
 Fws = .006       !m3/sec

 MG = 64.172      !Gal to m3 conversion
 MMW = 18.02      ! mol weight of H2O


 Global kf0, kr0, Ef0, Er0 As Real
 kf0 = 1.9E+5     !m3/mol sec
 kr0 = 5E+4       !m3/mol sec
 Ef0 = 59500      !J/mol
 Er0 = 59500      !J/mol

 Global V, Tr, rho, Cp, DHr, Tfeed, Tw, Cpw, Vc, rhoW, rhoWM, Tc As Real
 V = 10.0         !m3
 Tr = 343.15      !K
 rho = 17000      !mol/m3
 Cp = 100.0315    !J/mol K
 DHr= -29115      !J/mol
 Tfeed = 298.15   !K
 Tw = 298.15      !K
 Cpw=  75.4       !J/mol K
 rhoW = 55388.9   !mol/m3
 Vc = 0.1*5       !kg/m3
 rhoWM = 997      !kg/m3
 Tc = 321.5
```

```
@Initial Conditions
 U(1)=3.16305E+03
 U(2)=2.87822E+03
 U(3)=5.64904E+03
 U(4)=6.11190E+03  !mol/m3
 U(5)=3.49034E+02  !K
 !U(6)=3.21699E+02  !cooling water temp



@Model Equations
 Dim CAA, CEtoh, CEAC, CW, Temp As Real

 CAA = U(1)
 CEtoh = U(2)
 CEAC = U(3)
 CW = U(4)
 Temp = U(5)
 !Tc = U(6)



 Dim kfor, krev, R1 As Real

 !FcW = 36*sqrt(DP)*MG*rhoW*(1/MMW)*(1000/60)
 kfor= kf0*exp(-Ef0/(Rg*Temp))
 krev= kr0*exp(-Er0/(Rg*Temp))
 R1 = kfor*U(1)*U(2) - krev*U(3)*U(4)


 If(iRange==1) Tc = 321.5
 If(iRange==2) Tc = 331.5
 If(iRange==3) Tc = 311.5
 If(iRange==4) Tc = 325
```

```
F(1)=AAFeed - QF*CAA  -R1*V
F(2)=EtohFeed - QF*CEtoh  -R1*V
F(3)= -QF*CEAC + R1*V
F(4)= WFeed - QF*CW + R1*V
F(5)=QF*rho*Cp*(Tfeed - Temp) - Ua*(Temp-Tc) - DHr*R1*V
!F(6)=Fws*rhoW*Cpw*(Tw - Tc) + Ua*(Temp-Tc)

@Coefficient Matrix
E(1)=V
E(2)=V
E(3)=V
E(4)=V
E(5)=rho*Cp*V
!E(6)=rhoW*Cpw*Vc



@Solver Options
Neq=5
Npts=80
Tend=8000
Tstop=1000;2000;3000;4000
Param(28)=1
VariableNames=Time(Sec);CAA;CEtoh;CEAC;CW;Temp(K);Tc(K)
```

Feed Forward Code:

```
'Variabe Declarations and Data
'*******************************
 Parameter Rg=8.3145 As Real  !Universal Gas Constant, J/(mol K)

 Global AAFeed, EtohFeed, WFeed, QF, Ua, Qcool,DP, MG, MMW, Fws As Real
 AAFeed = 66.7148 !mol/sec
 EtohFeed = 64.5584     !mol/sec
 WFeed = 3.50421  !mol/sec
 QF = 0.007570824 !m3/sec
 Ua = 21586.3     !J/K sec
 Qcool = 431726   !J/mol
 DP = 7.26        !PSI
 Fws = .006       !m3/sec

 MG = 64.172      !Gal to m3 conversion
 MMW = 18.02      ! mol weight of H2O


 Global kf0, kr0, Ef0, Er0 As Real
 kf0 = 1.9E+5     !m3/mol sec
 kr0 = 5E+4       !m3/mol sec
 Ef0 = 59500      !J/mol
 Er0 = 59500      !J/mol

 Global V, Tr, rho, Cp, DHr, Tfeed, Tw, Cpw, Vc, rhoW, rhoWM As Real
 V = 10.0         !m3
 Tr = 343.15      !K
 rho = 17000      !mol/m3
 Cp = 100.0315    !J/mol K
 DHr= -29115      !J/mol
 Tfeed = 298.15   !K
 Tw = 298.15      !K
 Cpw=  75.4       !J/mol K
 rhoW = 55388.9   !mol/m3
 Vc = 0.1*5       !kg/m3
 rhoWM = 997      !kg/m3
```

```
@Initial Conditions
 U(1)=3.16305E+03
 U(2)=2.87822E+03
 U(3)=5.64904E+03
 U(4)=6.11190E+03  !mol/m3
 U(5)=3.49034E+02  !K
 U(6)=3.21699E+02  !cooling water temp


@Model Equations
 Dim CAA, CEtoh, CEAC, CW, Temp, Tc As Real

 CAA = U(1)
 CEtoh = U(2)
 CEAC = U(3)
 CW = U(4)
 Temp = U(5)
 Tc = U(6)


 Dim kfor, krev, R1 As Real

 !Fcw = 36*sqrt(DP)*MG*rhoW*(1/MMW)*(1000/60)
 kfor= kf0*exp(-Ef0/(Rg*Temp))
 krev= kr0*exp(-Er0/(Rg*Temp))
 R1 = kfor*U(1)*U(2) - krev*U(3)*U(4)


 If(iRange==1) Tw = 298.15
 If(iRange==2) Tw= 308.15
 If(iRange==3) Tw = 288.15
 If(iRange==4) Tw= 298.15
```

```
F(1)=AAFeed - QF*CAA  -R1*V
F(2)=EtohFeed - QF*CEtoh  -R1*V
F(3)= -QF*CEAC + R1*V
F(4)= WFeed - QF*CW + R1*V
F(5)=QF*rho*Cp*(Tfeed - Temp) - Ua*(Temp-Tc) - DHr*R1*V
F(6)=Fws*rhoW*Cpw*(Tw - Tc) + Ua*(Temp-Tc)

@Coefficient Matrix
E(1)=V
E(2)=V
E(3)=V
E(4)=V
E(5)=rho*Cp*V
E(6)=rhoW*Cpw*Vc



@Solver Options
Neq=6
Npts=80
Tend=8000
Tstop=1000;2000;3000;4000
Param(28)=1
VariableNames=Time(Sec);CAA;CEtoh;CEAC;CW;Temp(K);Tc(K)
```

Steady State Graphs

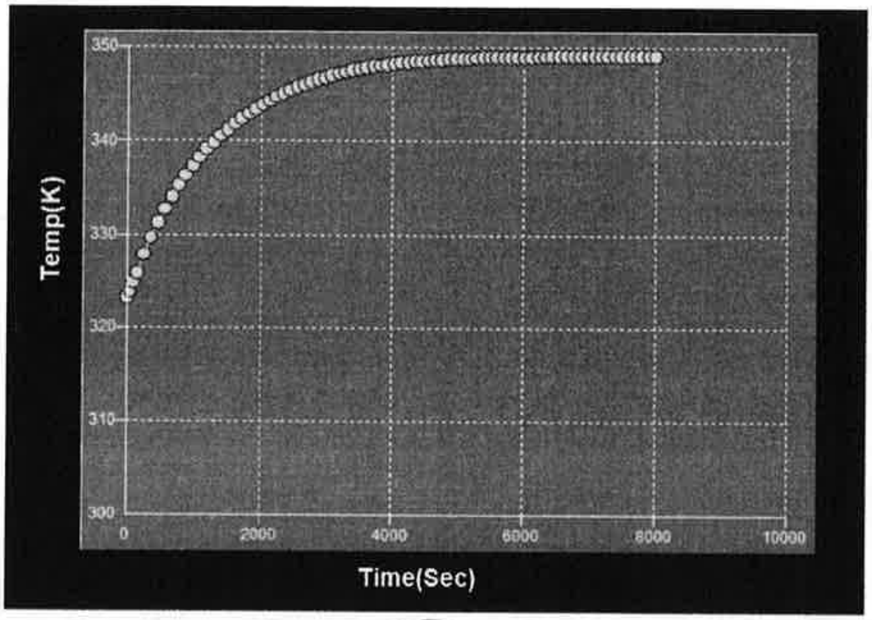

Figure 1: React temp vs time —1 Don't abbreviate here

what is Tc? —1

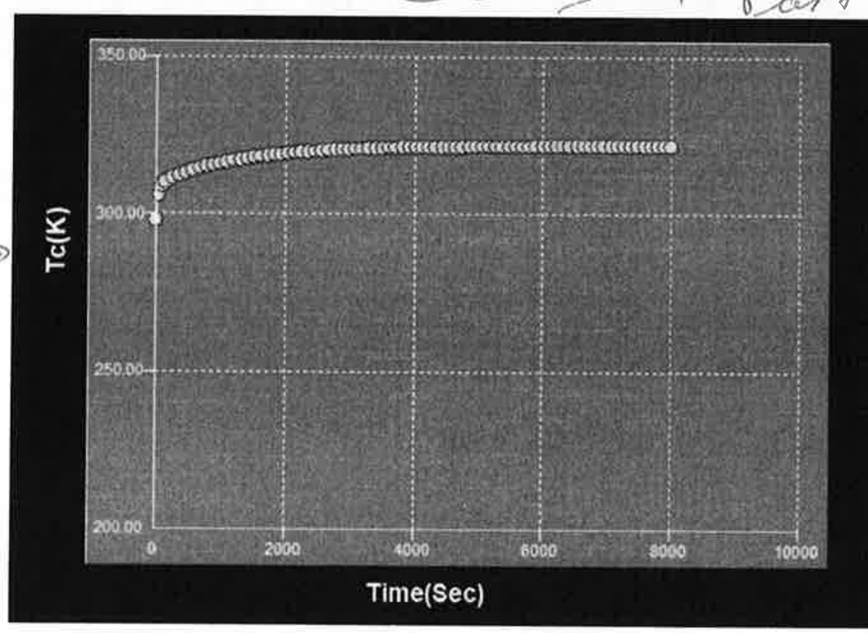

Figure 2: Cooling water temp vs time

2)

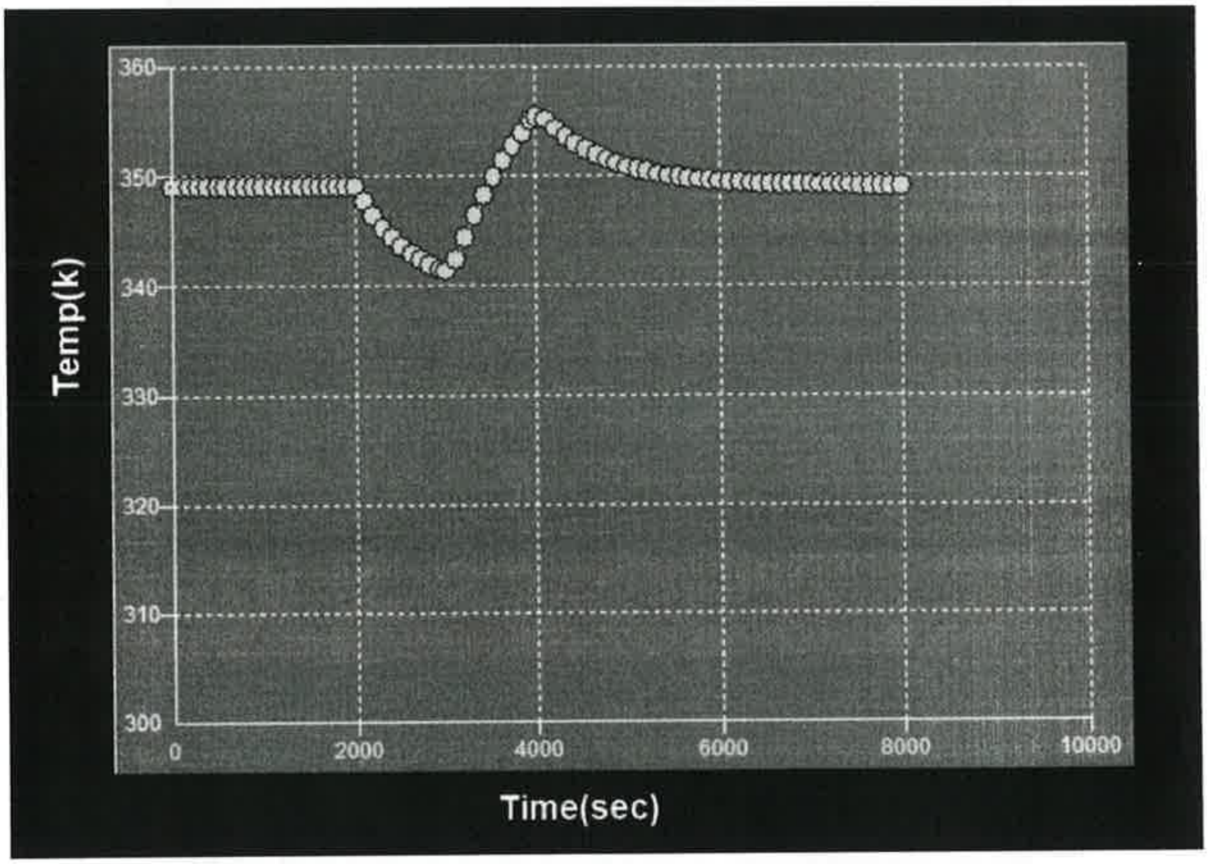

Figure 2: React temp vs time

Control variable?

-4

3)

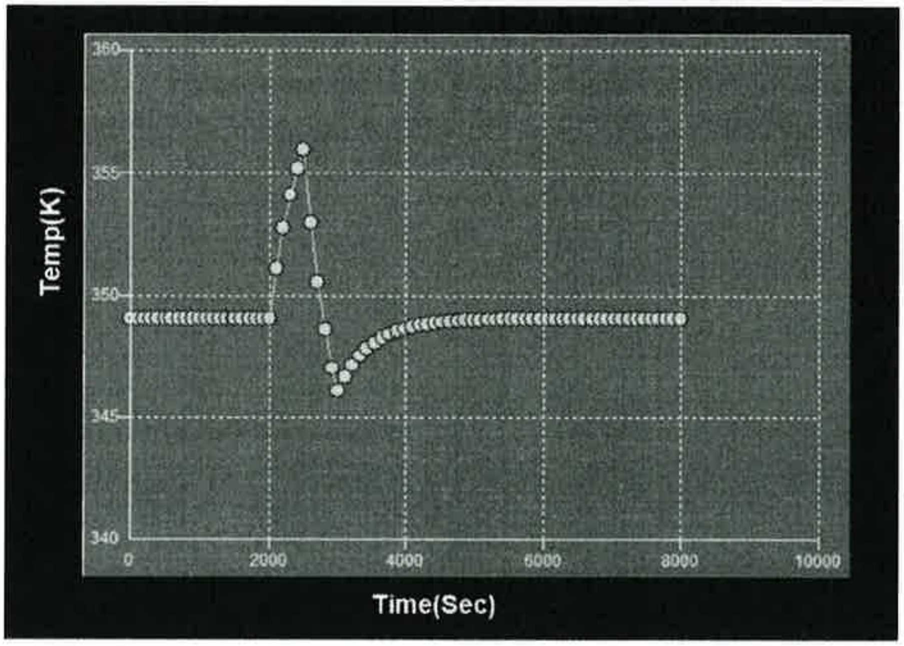

Figure 4: React temp vs time

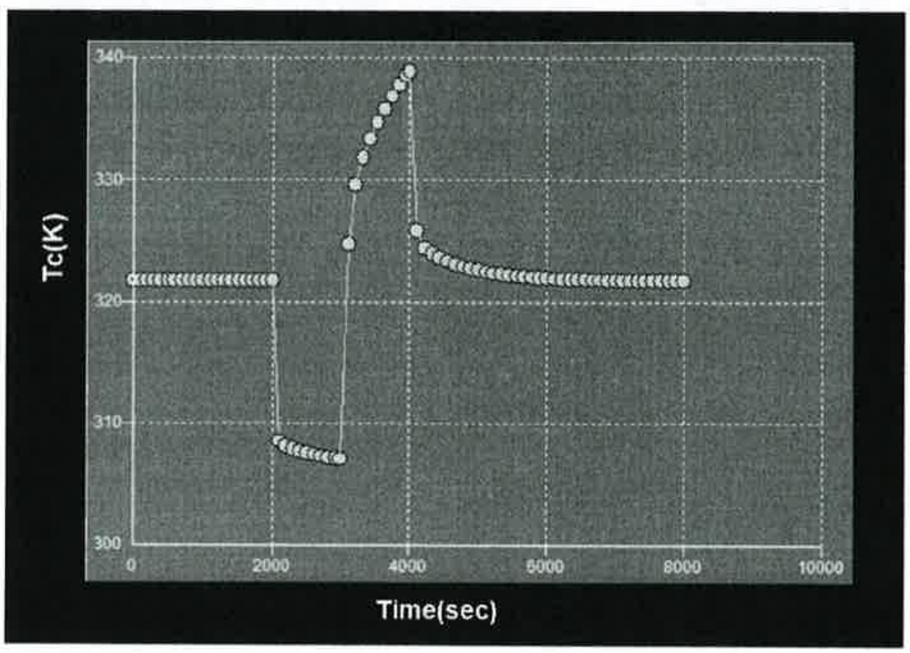

Figure 5: Cooling Jacket temp vs time

-3

4)

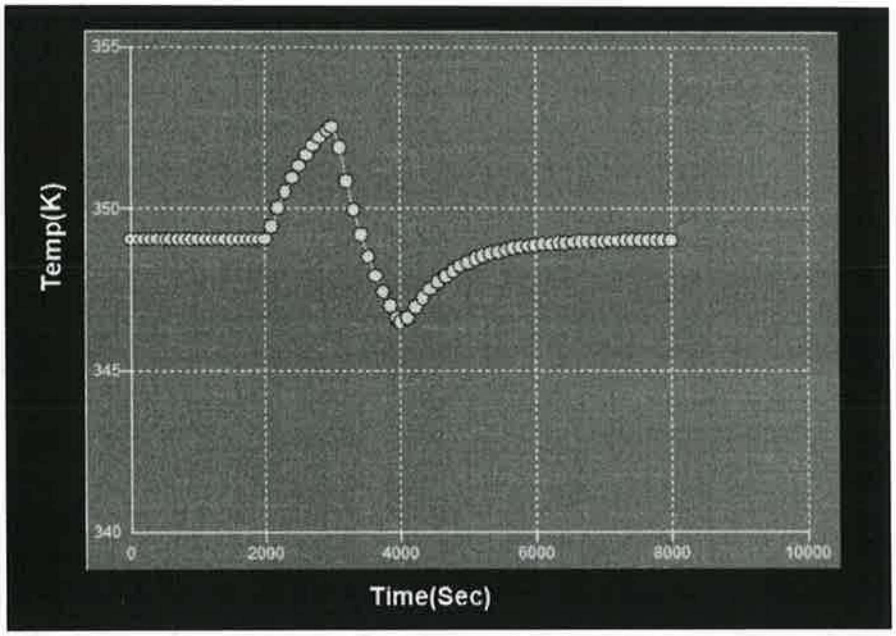

Figure 6: React temp vs time

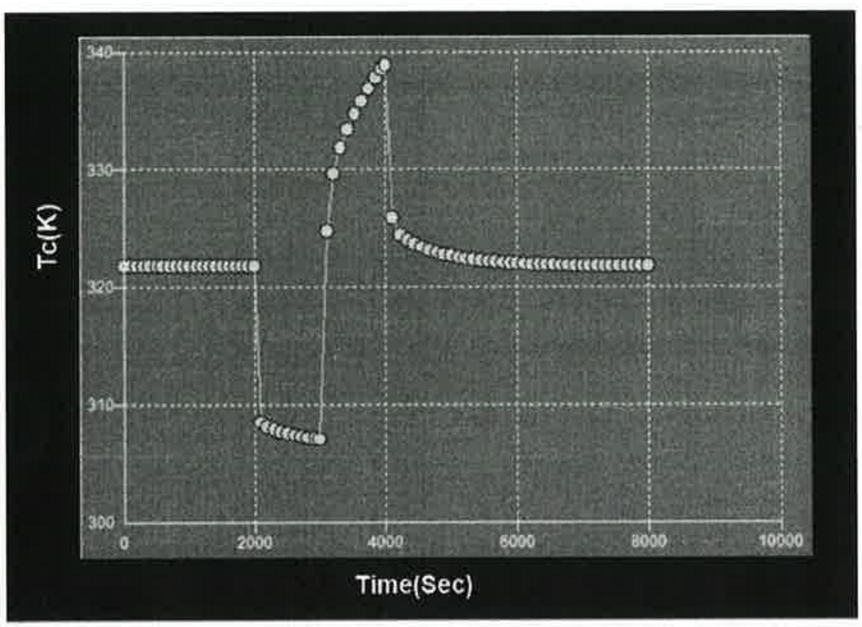

Figure 7: Cooling water temp vs time